

MEDTRACK_DV

Hospital Operations & Patient Analytics Dashboard

Infosys Springboard 7.0 Data Visualization (Batch -2)
Project Documentation

Implementation using Power BI and Tableau

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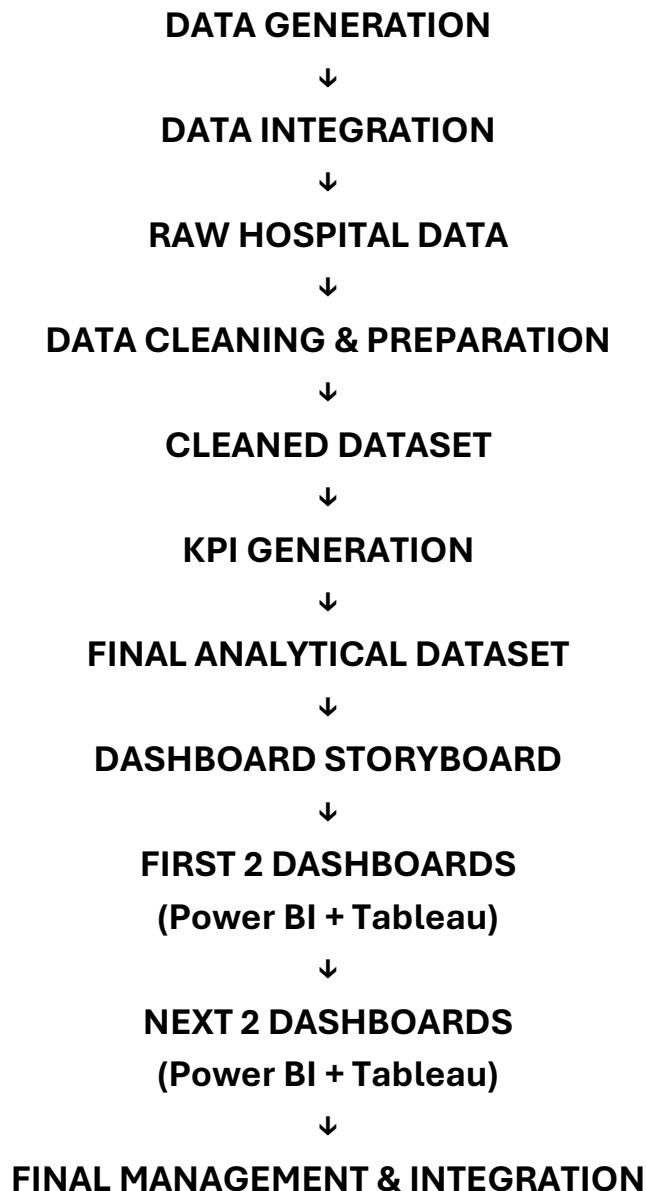
CMR COLLEGE OF ENGINEERING AND TECHNOLOGY
B.Tech / Data Science
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Project Overview

Hospital analytics project

MedTrack_DV is a hospital operations and patient analytics project completed as part of Infosys Springboard 7.0. The project takes hospital data through a structured process and converts it into useful KPIs and interactive dashboards. The dashboards were developed in both Power BI and Tableau.

The project was completed progressively through seven modules. Each module produced an output that was used in the following stage, creating a clear end-to-end workflow from raw data to final tested dashboards.



The four-dashboard scope covers Hospital Overview, Patient Flow, Department Analytics and Resource Utilization. The final stage consolidates the work and verifies that the dashboards and their interactions operate correctly.

Problem Statement

Hospitals generate information from many operational activities, including patient admissions, patient movement, department performance, bed occupancy, staffing and other resources. When these measures are viewed separately, it can be difficult to understand the overall operational situation.

A dashboard solution can bring the important measures into one analytical environment. It can help users compare performance, observe trends and identify areas that may require attention.

How MedTrack_DV addresses the problem

- Combines the project data into a structured analytical dataset.
- Cleans and prepares the data before visualization.
- Calculates important hospital performance indicators.
- Presents information through interactive Power BI and Tableau dashboards.
- Separates the analysis into four clear operational areas.
- Provides a final managed and tested dashboard solution.

Expected Value

The project provides a simple visual way to understand hospital operations. Instead of depending only on raw tables, users can use KPIs, charts, filters and dashboard interactions to explore the available information.

Project Objectives

The project objectives were organized around the complete analytics lifecycle:

Objective	Description
Data preparation	Generate, integrate, clean and prepare hospital data.
KPI development	Create useful hospital performance measures from the prepared data.
Dashboard planning	Prepare a storyboard before dashboard implementation.
Dashboard development	Build four dashboards covering the main hospital analytics areas.
Dual-platform implementation	Develop the dashboard solution in both Power BI and Tableau.
Integration	Bring the dashboard work into final project files.
Testing	Check calculations, filters, navigation and dashboard behavior.
Documentation	Maintain a clear project record and deliver the work through GitHub.

Project Outcome

At completion, the project contains the required data-processing files, datasets, storyboard, Power BI files, Tableau workbooks and final documentation. The implementation follows a module-based workflow so that each stage can be traced from its input to its output.

Tools & Technologies

Technology	Use in the Project
Python	Used for data generation, integration and KPI-related processing.
Pandas / NumPy	Used where required for data preparation and transformation.
Jupyter Notebook	Used for the data-cleaning stage.
CSV	Used for raw and cleaned hospital datasets.
Excel	Used for the final analytical dataset.
Power BI	Used to create and manage the hospital dashboards.
Tableau	Used to create and manage the hospital dashboards.
GitHub	Used to store, organize and deliver the project.
PDF	Used for the Module 4 dashboard storyboard.

Why two BI platforms were used

Power BI and Tableau were both used for dashboard implementation. Separate prototype files were maintained during Modules 5 and 6, and final files were maintained in Module 7. This gives the project two complete visualization implementations while keeping the underlying analytical objective consistent.

Module 1 – Data Generation & Integration

Module 1 established the data foundation of the project. The work in this stage focused on creating the hospital data and integrating the required information into a raw dataset that could be cleaned in the next module.

File	Purpose
data_generation.py	Python script used for hospital data generation/preparation.
data_integration.py	Python script used to integrate the required data.
hospital_raw_data.csv	Raw hospital dataset produced for further processing.

Module 1 Output

The main output was hospital_raw_data.csv. This raw dataset became the input for the data-cleaning process in Module 2.

Module 2 – Data Cleaning & Preparation

Module 2 converted the raw hospital data into a cleaner dataset suitable for analytics. The cleaning work was performed in a Jupyter Notebook and the resulting data was saved separately so that the original raw data remained available.

File	Purpose
data_cleaning.ipynb	Notebook containing the data-cleaning and preparation process.
hospital_cleaned.csv	Cleaned hospital dataset used in later modules.

Main Preparation Activities

- Review the raw data for quality issues.
- Handle missing or unsuitable values where required.
- Remove duplicate records where applicable.
- Standardize relevant categorical values such as department information.
- Prepare consistent fields for KPI generation and dashboard analysis.

Module 2 Output

The cleaned CSV became the reliable input for the KPI-generation stage. Keeping a separate cleaned file also makes the project workflow easier to understand and reproduce.

Module 3 – KPI Engineering

Module 3 transformed the cleaned information into a final analytical dataset and generated the hospital KPIs required for dashboard analysis.

File	Purpose
generate_hospital_kpis.py	Python script used for hospital KPI generation.
hospital_final_dataset.xlsx	Final analytical dataset prepared for dashboard development.

Main KPIs

KPI	Meaning
Total Admissions	Total number of patient admissions in the available data.
Occupancy Rate	Indicates the level of hospital occupancy.
Average Length of Stay	Shows the average duration of patient stay.
Readmission Rate	Indicates the level of patient readmission.
Bed Utilization Rate	Shows how available hospital beds are being used.
Department Efficiency Score	Supports comparison of department performance.

These KPIs provide the numerical foundation for the visual analysis presented in the dashboards.

Module 4 – Dashboard Storyboard

Module 4 was a planning stage rather than a dashboard-development stage. The main work completed in this module was the creation of the dashboard storyboard PDF.

The storyboard helped define the intended dashboard structure, visual organization and analytical flow before the actual dashboard files were developed in Power BI and Tableau.

Deliverable	Description
dashboard_storyboard.pdf	PDF containing the dashboard storyboard / planning work.

Role in the Project

- Provides a visual plan before dashboard implementation.
- Helps organize the dashboard sections and information.
- Acts as a reference during Power BI and Tableau development.
- Connects the data/KPI stages with the dashboard-development stages.

Module 5 – First Two Dashboards

Module 5 was the first dashboard-development stage. The first two dashboards were implemented in both Power BI and Tableau. Separate prototype files were maintained for the two platforms.

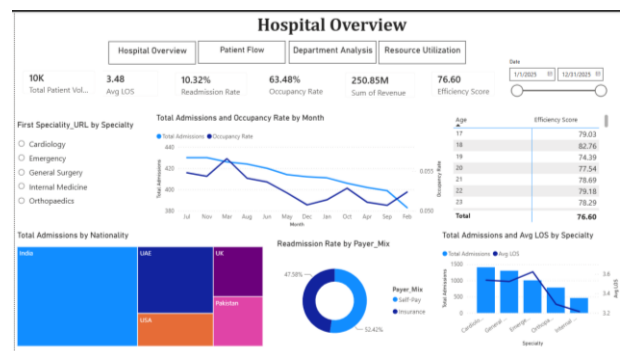
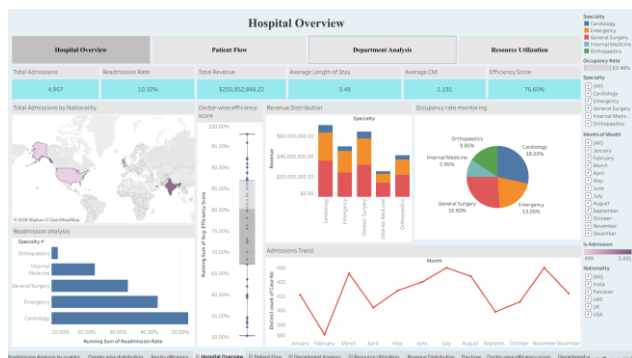
Platform	File
Power BI	medtrack_prototype_1&2.pbix
Tableau	medtrack_prototype_1&2.twb

Development Focus

The first two dashboards form the initial dashboard set of the project. They were developed using the prepared data and KPI foundation from the previous modules.

What to document with screenshots

- Dashboard name and purpose.
- Main KPIs displayed.
- Important charts or visual elements.
- Filters or interactive controls.
- One or two useful observations from the dashboard.



Module 6 – Next Two Dashboards

Module 6 continued the dashboard-development process by creating the next two dashboards. These dashboards were also implemented in both Power BI and Tableau.

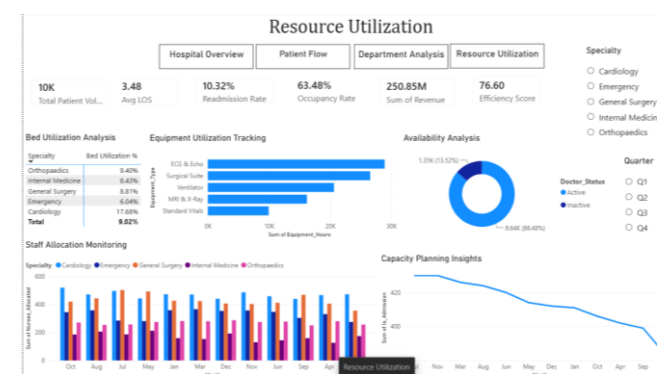
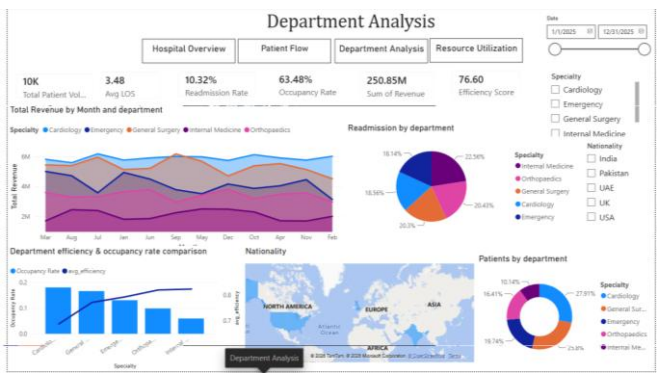
Platform	File
Power BI	medtrack_prototype_3&4.pbix
Tableau	medtrack_prototype_3&4.twb

Development Focus

With the completion of Module 6, the project reaches the planned four-dashboard scope. The dashboards together provide a broader view of hospital operations rather than focusing on a single measure.

Four Dashboard Areas

Dashboard	Focus
Hospital Overview	Overall hospital performance and major KPIs.
Patient Flow	Admissions, discharges, patient movement and stay-related analysis.
Department Analytics	Department-level volume, performance and efficiency.
Resource Utilization	Beds, staff, equipment and operational resource usage.



Module 7 – Final Management, Integration & Testing

Module 7 is the final project stage. The dashboard work from Modules 5 and 6 was brought into final project files for complete management, integration and testing.

Item	Final File
Power BI	medtrack_prototype.pbix
Tableau	medtrack_prototype.twb
Project documentation	README.md

Final Activities

- Bring the completed dashboard work into final project files.
- Check the consistency of the final dashboard solution.
- Verify KPIs and important calculations.
- Test filters, navigation and interactive actions where implemented.
- Check the final Power BI report and Tableau workbook.
- Confirm that the project files and documentation are organized in GitHub.

Testing & Validation

Testing was completed as part of Module 7 to ensure that the final dashboard solution was ready for use and delivery. The testing stage should cover both the analytical values and the interactive behavior of the dashboards.

Test Area	Validation
KPI calculations	Check that displayed KPI values are consistent with the prepared data.
Data consistency	Check that the correct final dataset is being used.
Filters	Confirm that filters affect the intended charts and KPIs.
Navigation	Confirm that navigation works where it is implemented.
Dashboard actions	Check interactive actions and selections.
Power BI report	Open the final PBIX and verify visuals and interactions.
Tableau workbook	Open the final TWB and verify dashboards and interactions.
Final review	Check layout, labels, readability and overall usability.

Results & Key Insights

The completed MedTrack_DV project provides a structured hospital analytics solution from raw data through final interactive dashboards. The implementation covers four operational areas and provides both Power BI and Tableau versions.

Project Results

- Hospital data was generated and integrated into a raw dataset.
- The raw dataset was cleaned and prepared for analytics.
- Hospital KPIs were generated and included in the final analytical dataset.
- A dashboard storyboard was prepared before implementation.
- Four dashboard areas were developed across Power BI and Tableau.
- Final project files were created and managed in Module 7.
- The completed solution was tested before documentation and delivery.

Key Insights from the Dashboards

Add 2–4 actual observations from each dashboard here. Use only values and trends visible in your completed dashboards. Do not add made-up numbers. A good insight should state what was observed and what it means operationally.

Dashboard	Insights to Add
Hospital Overview	Overall admissions/KPI trends and major hospital-level observations.
Patient Flow	Important admission, discharge, stay or patient-movement trends.
Department Analytics	Strongest department comparisons or efficiency observations.
Resource Utilization	Important bed, staff, equipment or capacity observations.

GitHub Repository & Project Structure

The project is maintained in GitHub under the repository Hospital-Operations-and-Patient-Analytics-Dashboard-Group-2, with the work organized under the Nihal-Kata branch.

Folder	Important Contents
MODULE 1	data_generation.py, data_integration.py, hospital_raw_data.csv
MODULE 2	data_cleaning.ipynb, hospital_cleaned.csv
MODULE 3	generate_hospital_kpis.py, hospital_final_dataset.xlsx
MODULE 4	dashboard_storyboard.pdf
MODULE 5 / POWER BI	medtrack_prototype_1&2.pbix
MODULE 5 / TABLEAU	medtrack_prototype_1&2.twb
MODULE 6 / POWER BI	medtrack_prototype_3&4.pbix
MODULE 6 / TABLEAU	medtrack_prototype_3&4.twb
MODULE 7 / POWER BI	medtrack_prototype.pbix
MODULE 7 / TABLEAU	medtrack_prototype.twb
Root	README.md

Repository Link

<https://github.com/springboardmentorpl/Hospital-Operations-and-Patient-Analytics-Dashboard-Group-2/tree/Nihal-Kata>

Conclusion

MedTrack_DV was completed as an end-to-end hospital operations and patient analytics project under Infosys Springboard 7.0. The implementation follows a clear progression from data preparation to business intelligence visualization.

Module 1 created and integrated the hospital data. Module 2 cleaned the data, while Module 3 generated the analytical KPIs and final dataset. Module 4 was used for dashboard storyboard planning. Module 5 developed the first two dashboards and Module 6 developed the next two dashboards, with both stages implemented in Power BI and Tableau. Module 7 brought the work into final files for complete management, integration and testing.

The final project demonstrates how structured data preparation, KPI engineering and interactive visualization can be combined to present hospital operational information in a clear and useful form.